TotalEnergies

TotalEnergies Fluids

TEMAP Distributor Seminar

Xiamen, March 2026

Key data on Special Fluids

Our ambition

A successful pioneer in pure and sustainable Fluids

- Value approach: optimize portfolio to increase value
- Sustainability : 30 % low carbon Fluids by 2030
- Simplified and resilient business and operations

Our figures

- Sales: 600 KT HDA – 200 KT aromatics & others
- N°3 worldwide, joint leader in Europe
- 26 M$ de Cash Flow en 2025
- 500 customers represent 80% of our sales
- 2 plants: Oudalle (France) and Bayport (USA)
- A global network: 60 countries

Our HDA Special Fluids offer

- A range of HIGH PURITY products
- Best-in-class low aromatics content
- BioLife renewable range and EcoLife sustainable offer

100 years of Special Fluids

- A business fully integrated into the Company since 1989
- A continuous adaption of our industrial tool to the market demand

1921 – **Creation** of "Société de Droguerie et d'Hydrocarbures de Saint-Denis"

1928 – **The company is renamed HSD** « Société d'Hydrocarbures de Saint-Denis »

1955 – HSD acquires **a distillation unit**

1968 – **Arson** at the Saint-Denis factory **Decision to transfer production** to near the Normandy Refinery

1969-1970 – **Construction of the Oudalle plant**

1989 – Total takes 100% of HSD, and becomes **Total Solvents**

1995 – **Construction of the hydrogenation units** Hydro 1, 2 et 3

2004 – **Total Solvents becomes Total Fluides** Construction of a hydrogenation unit Hydro 4

2008 – Construction of **2 Blast Proof buildings** Renovation of the laboratory

2016 – **Start-up of the site in Bayport** (United States)

2021 – **Total Fluides becomes TotalEnergies Fluids**

Our roadmap

Sustainable Growth & Expansion

Customer Proximity and Operational Efficiency

Commitment and Skills

To achieve this roadmap

Product Development & Regulation Department

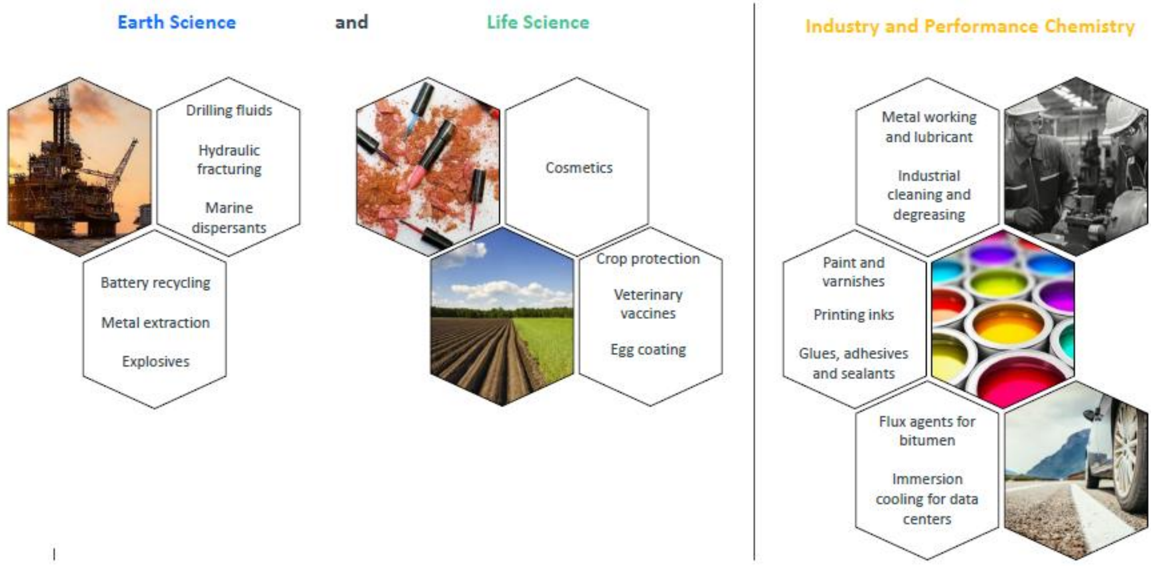
- Definition of the **Research and Development roadmap** as well as the **Regulation roadmap**.
- Responsible for the **Special Fluids range** Manages **technical** product and analytical **support**
- **R**esponsible for developing **knowledge** on Special Fluids products
- Responsible for certification, registration, or approval requests for products with the relevant authorities
- Manages **the patent portfolio** of the Special Fluids Division with the relevant support entities.

Each Business Line is responsible for

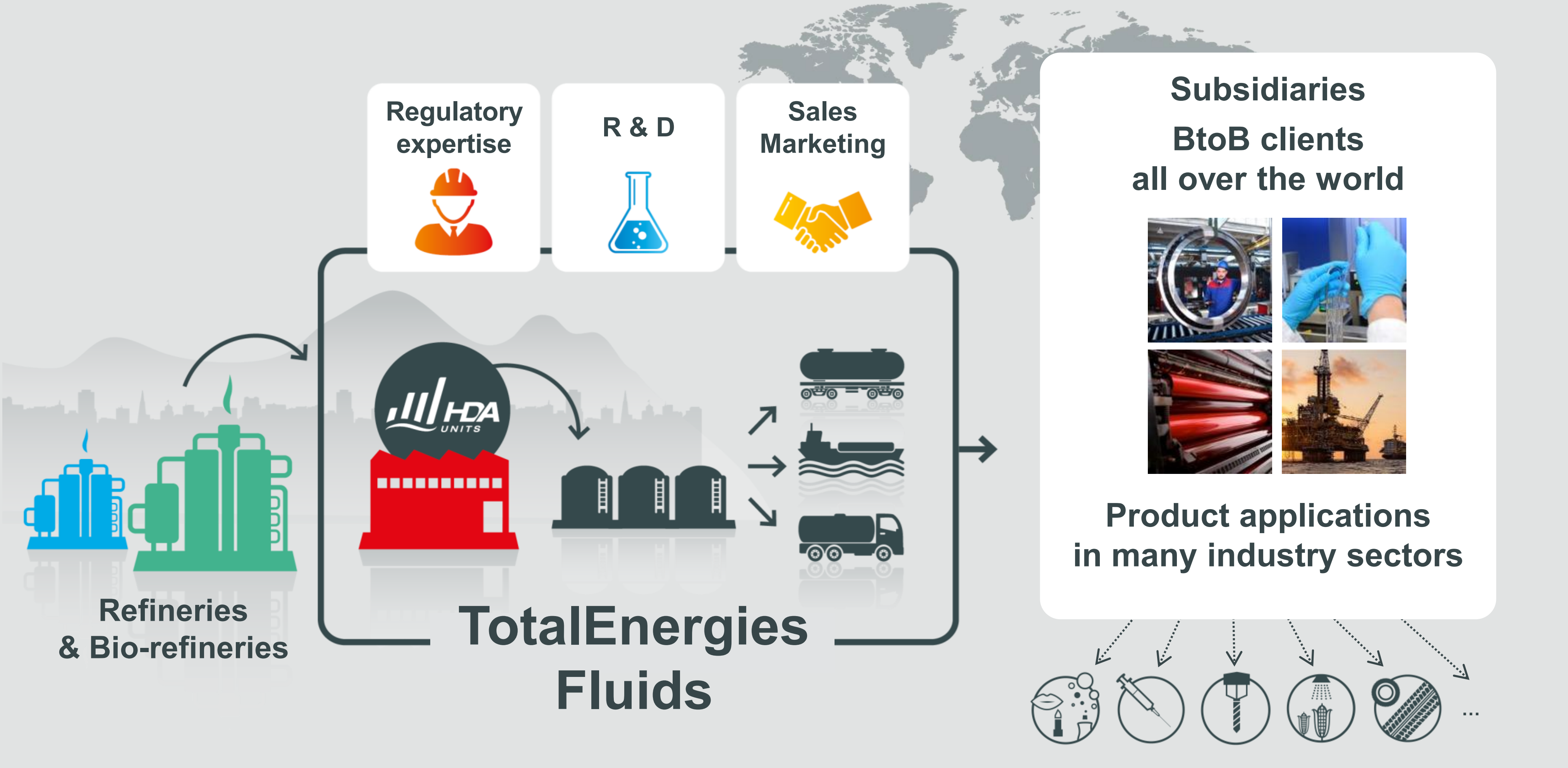
- The knowledge of the markets at the global level
- The definition of the product range
- The definition of the commercial & marketing strategy
- The support to the zones

2 Types of Jobs in each BL

- Key account Managers
- Market Managers



So… the Special Fluids in a nutshell



01.

TotalEnergies Fluids: Our process

Our HDA UNITS process: a know-how that sets us apart

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Gasoline

Jet

Diesel

Bio feedstocks

- **High-pressure catalytic hydrogenation** (50-150 bars H_2)
- Capacity of treating a broad range of feedstocks

Multiple distillations

- Narrow cuts: from 20° to 70°C
- Flexibility

Dearomatized special fluids

Aromatics < 300 ppm

PAH* < 100 ppb

The highest level of purity in the market

*PAH: Polycyclic Aromatic Hydrocarbon

- **Our patented HDA technology allows us to obtain the purest products in the market.**
- **Our plants stand out by their capacity to process diesel feedstocks and to manufacture specialty products.**

02.

Our sustainable ranges

BioLife: From Bio-circular to pure Hydrocarbons

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Our processes

From Bio-circular materials to pure Hydrocarbons

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Traceability and Certifications

From bio-circular raw materials

ISCC PLUS certified process. Traceability throughout the whole production chain

As a result of carbon dioxide (CO_2) capture during plant growth, the net greenhouse gas (GHG) impact is negative. Biogenic sequestration outweighs the emissions generated during harvesting and processing. Carbon footprint data is available for all products.

BioLife: From Bio-circular to pure Hydrocarbons

Purity

Pure Hydrocarbons

High iso-paraffinic content (>80%)

Extra low aromatics levels (< 20 ppm)

Readily biodegradable (according to OECD 306 / 301)

BioLife: From Bio-circular to pure Hydrocarbons

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Performance

Excellent oxidation and thermal stability	High Flash point – Safe use
Narrow boiling ranges – Low volatility	Transparency
Very low Pour point	Odorless

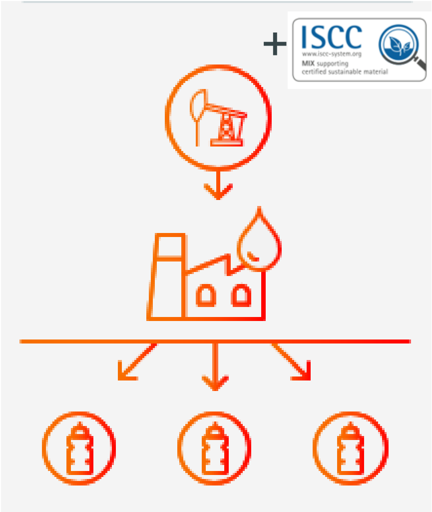
Choose your path

Traceability and Certifications

BioLife

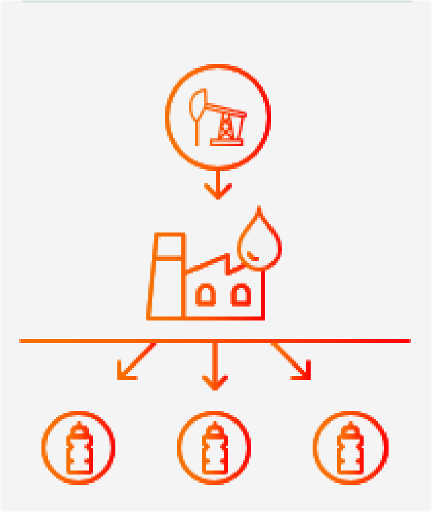
100% bio-based
Carbon negative
ISCC+ certified

EcoLife



Fossil based
Carbon neutral
ISCC+ certified

Standard



Fossil based
Carbon positive

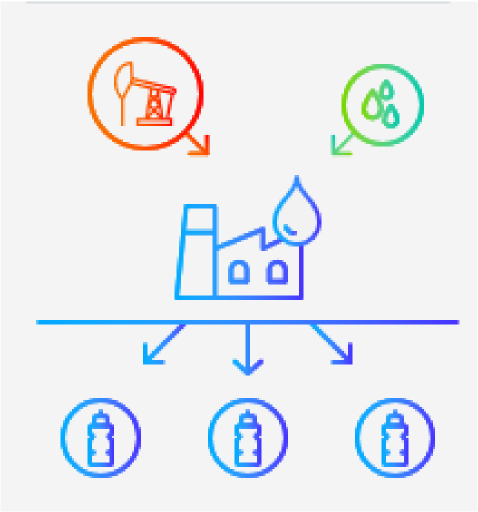
Choose your path

Traceability and Certifications

BioLife

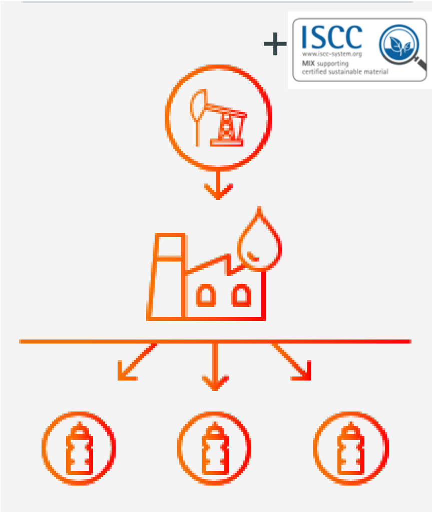
100% bio-based
Carbon negative
ISCC+ certified

NeutraLife



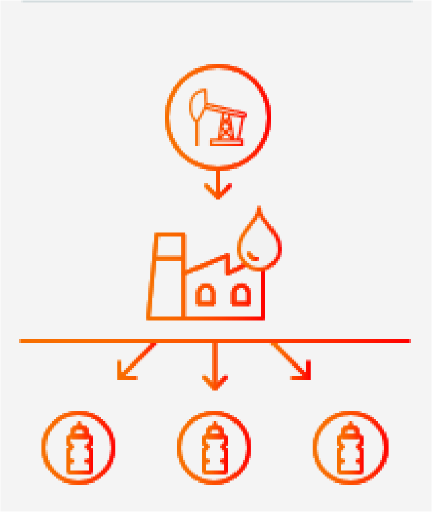
Partially bio-based
Carbon neutral
ISCC+ certified

EcoLife



Fossil based
Carbon neutral
ISCC+ certified

Standard



Fossil based
Carbon positive

Our Priority markets

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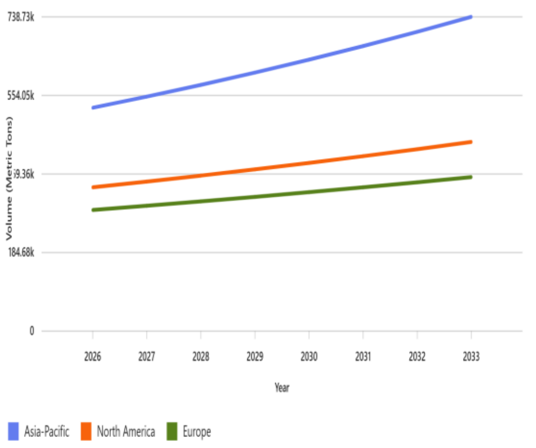
Sealants

Global sealants market in construction sector
1 330 KT
(+4% growth per year)

AP: 550 kt (CAGR +5%)
NA: 360 kt (CAGR +4,5%)
EUR: 300 kt (CAGR +3,5%)
SA: 60 kt (CAGR +3%)



Lubricants

Market forecast

2025 .
Global industry: **14 MT**
MWF: **1, 8 MT**

2025-2030 CAGR
Global industry : **+0,96%**
MWF: **+0,90%**

03.

Product overview

Generic ranges: Multi applications

Special Boiling point solvents

ALIPHATIC HYDROCARBON SOLVENTS

Typical values

PROPERTIES	UNITS	METHODS	SOLANE 60-95	SOLANE C	SOLANE 80-110	SOLANE 100-140 HN	SOLANE 100-160
DENSITY AT 15°C (59°F)	kg/m³	ISO 12185	676	704	695	731	765
SAYBOLT COLOR	Saybolt color unit	ASTM D156	+30	+30	+30	+30	+30
BOILING RANGE: INITIAL POINT	°C (°F)	ASTM D86	62 (144)	71 (160)	84 (183)	102 (216)	124 (255)
FINAL POINT	°C (°F)	ASTM D86	92 (198)	89 (192)	107 (225)	138 (280)	157 (315)
FLASH POINT: ABEL	°C (°F)	ISO 13736	< -35 (< -31)	-30 (-31)	-22 (-8)	-7 (18)	14 (57)
EVAPORATION RATE	Ether = 1	DIN 53170	2	2	3	6	11
	nBuAc = 1	ASTM D3539	0.2	0.3	0.3	0.5	1
AROMATIC CONTENT	mg/kg (ppm)	UV Internal method	< 20	< 20	< 20	< 20	<20
N-HEXANE CONTENT	% masse	GC Internal method	1.1	30	0.2	0.03	0.03
BENZENE CONTENT	mg/kg (ppm)	ASTM D6229	< 1	< 1	< 1	< 1	<1
SULPHUR CONTENT	mg/kg (ppm)	ASTM D5453	< 1	< 1	< 1	< 1	<1
ANILINE POINT	°C (°F)	ASTM D611	70 (158)	57 (135)	68 (154)	61 (142)	59 (138)
PARAMETERS HANSEN	Dispersion force D (MPa$^{1/2}$)		16.40	16.40	14.20	-	-
	Polar force P (MPa$^{1/2}$)		2.23	2.23	4.22	-	-
	Hydrogen bonding force H (MPa$^{1/2}$)		4.10	4.10	0.72	-	-
VAPEUR PRESSURE AT 20°C (68°F)	kPa	Calculated	14	13	7	3	1
VISCOSITY AT 20°C (68°F)	mm²/s	ASTM D445	0.5	0.6	0.6	0.7	1
CARBON CHAIN LENGTH			C6 - C7	C6 - C7	C7	C7 - C9	C7 - C9
C.A.S. NUMBER			64742-49-0 (R)	64742-49-0 (R)	64742-49-0 (R)	64742-49-0 (R)	64742-48-9 (R)
EC NUMBER			921-024-6	924-168-8	927-510-4	920-750-0	920-750-0

- Adhesives
- Aerosols
- Demoulding agents
- Polymerisation
- Tires

Hexane and Heptanes

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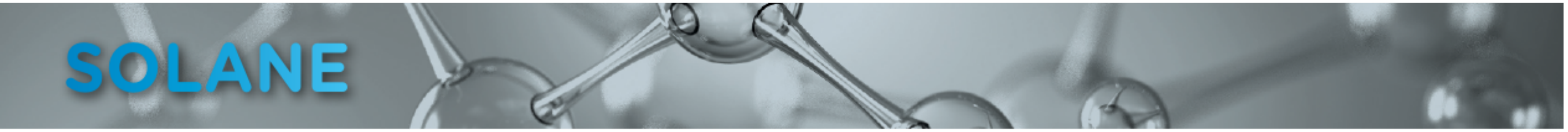


ALIPHATIC HYDROCARBON SOLVENTS C6 TO C7

Typical values

PROPERTIES	UNITS	METHODS	SOLANE ISOHEXANE	SOLANE HEXANE	SOLANE HEXANE 45	SOLANE HEPTANE
DENSITY AT 15°C (59°F)	kg/m³	ISO 12185	662	675	670	694
SAYBOLT COLOR	Saybolt color unit	ASTM D156	+30	+30	+30	+30
BOILING RANGE: INITIAL POINT	°C (°F)	ASTM D1078	55 (131)	63 (145)	65 (149)	90 (194)
FINAL POINT	°C (°F)	ASTM D1078	61 (142)	71 (160)	69 (156)	94 (201)
FLASH POINT: ABEL	°C (°F)	ISO 13736	< -35 (< -31)	< -35 (< -31)	< -35 (< -31)	-31 (-24)
EVAPORATION RATE	Ether = 1	DIN 53170	1	2	2	3
EVAPORATION RATE	nBuAc = 1	ASTM D3539	0.2	-	-	0.3
AROMATIC CONTENT	mg/kg (ppm)	UV Internal method	< 20	30	< 20	< 20
N-HEXANE CONTENT	% masse	GC Internal method	1.9	45	48	0.1
BENZENE CONTENT	mg/kg (ppm)	ASTM D6229	< 1	< 1	< 1	< 1
SULPHUR CONTENT	mg/kg (ppm)	ASTM D5453	< 1	< 1	< 1	< 1
ANILINE POINT	°C (°F)	ASTM D611	75 (167)	70 (158)	71 (160)	77 (171)
PARAMETERS HANSEN	Dispersion force δD (MPa$^{1/2}$)		-	-	-	15.3
PARAMETERS HANSEN	Polar force δP (MPa$^{1/2}$)		-	-	-	0
PARAMETERS HANSEN	Hydrogen bonding force δH (MPa$^{1/2}$)		-	-	-	0
VAPEUR PRESSURE AT 20°C (68°F)	kPa	Calculated	26	18	18	7
VISCOSITY AT 20°C (68°F)	mm²/s	ASTM D445	0.45	0.47	0.47	0.57
BROMINE INDEX	mg Br / 100 g	ASTM D2710	-	0.2	0.2	0.1

	SOLANE ISOHEXANE	SOLANE HEXANE	SOLANE HEXANE 45	SOLANE HEPTANE
CARBON CHAIN LENGTH HSAP	C6	C6	C6	C7
C.A.S. NUMBER	64742-49-0 (R)	92112-69-1 (R)	92112-69-1 (R)	64742-49-0 (R)
EC NUMBER	931-254-9	925-292-5	925-292-5	927-510-4

- Adhesives
- Aerosols
- Demoulding agents
- Polymerisation
- Resins

SOLANE HEXANE 45

Vegetable oil extraction

Cyclic solvents

CYCLO-ALIPHATIC HYDROCARBON SOLVENTS

Typical values

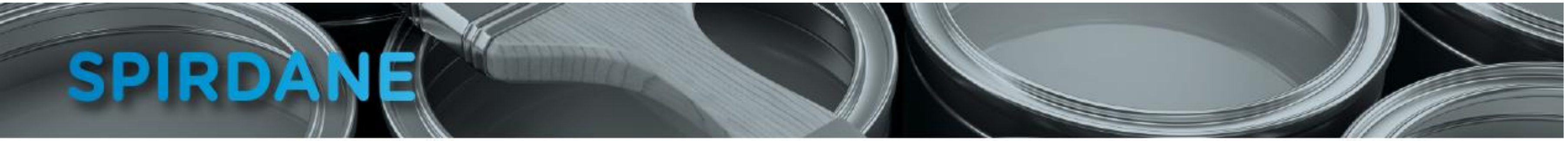
PROPERTIES	UNITS	METHODS	SOLANE CYCLOPENTANE	SOLANE CYCLOHEXANE	SOLANE METHYLCYCLOHEXANE
DENSITY AT 15°C (59°F)	kg/m³	ISO 12185	748	782	774
COLOR PT/CO	Hazen	ASTM D1209	6.5	5.8	< 5
BOILING RANGE: INITIAL POINT	°C (°F)	ASTM D86	49 (120)	-	101 (214)
FINAL POINT			50 (122)	-	102 (216)
RANGE (PI/PF)		ASTM D 1078	-	0.1 (32.2)	-
FLASH POINT: ABEL	°C (°F)	ISO 13736	< -35 (< -31)	< - 15 (< 5)	< - 13 (< 9)
EVAPORATION RATE	Ether = 1	DIN 53170	1	2	3
	nBuAc = 1	ASTM D3539	-	0.5	0.5
AROMATIC CONTENT	mg/kg (ppm)	UV internal method	< 20	< 20	< 20
CYCLOPARAFFIN CONTENT	% masse	GC - MS	97.8	99.98	99.8
BENZENE CONTENT	mg/kg (ppm)	ASTM D6229	< 1	< 1	< 1
SULPHUR CONTENT	mg/kg (ppm)	ASTM D5453	< 1	< 1	< 1
ANILINE POINT	°C (°F)	ASTM D611	20 (68)	30 (86)	39 (102)
PARAMETERS HANSEN	Dispersion force D (MPa$^{1/2}$)		-	16.8	16
	Polar force P (MPa$^{1/2}$)		-	0	0
	Hydrogen bonding force H (MPa$^{1/2}$)		-	0.2	1
VAPEUR PRESSURE AT 20°C (68°F)	kPa	Calculated	35	10	5
VISCOSITY AT 20°C (68°F)	mm²/s	ASTM D445	0.6	1.3	0.95

	SOLANE CYCLOPENTANE	SOLANE CYCLOHEXANE	SOLANE METHYLCYCLOHEXANE
CARBON CHAIN LENGTH HSAP	C5	C6	C7 - C8
C.A.S. NUMBER	287-92-3	110-82-7	108-87-2 (R)
EC NUMBER	206-016-6	203-806-2	927-033-1

SOLANE MCH

- Adhesives
- Resins
- Pharma
- Tires

Dearomatized white spirits



DEAROMATIZED WHITE SPIRITS – FAST EVAPORATION

Typical values

PROPERTIES	UNITS	METHODS	SPIRDANE D 25	SPIRDANE D 30	SPIRDANE D 40	SPIRDANE D 60
DENSITY AT 15°C (59°F)	kg/m³	ASTM D4052	769	779	789	807
SAYBOLT COLOR	Unit Saybolt color	ASTM D6045	+30	+30	+30	+30
BOILING RANGE: INITIAL POINT	°C (°F)	ASTM D86	140 (284)	149 (300)	156 (313)	185 (365)
FINAL POINT			161 (322)	171 (340)	199 (390)	214 (417)
FLASH POINT: PM	°C (°F)	ASTM D93	-	-	45 (113)	66 (151)
ABEL		ISO 13736	25 (77)	32 (90)	-	-
EVAPORATION RATE	Ether=1	DIN 53170	13	17	35	78
AROMATIC CONTENT	mg/kg (ppm)	UV Internal method	< 20	< 20	< 20	< 20
BENZENE CONTENT	mg/kg (ppm)	ASTM D6229	< 1	< 1	< 1	< 1
SULPHUR CONTENT	mg/kg (ppm)	ASTM D5453	< 1	< 1	< 1	< 1
ANILINE POINT	°C (°F)	ASTM D611	60 (140)	62 (144)	63 (145)	66 (151)
VAPOR PRESSURE AT 20°C (68°F)	kPa	Calculated	0.59	0.34	0.16	0.09
SURFACE TENSION AT 20°C (68°F)	mN/m	NF EN 14370	23.2	23.5	23.7	25.1
VISCOSITY AT 20°C (68°F)	mm²/s	ASTM D445	1.1	1.2	1.4	1.9

	SPIRDANE D 25	SPIRDANE D 30	SPIRDANE D 40	SPIRDANE D 60
CARBON CHAIN LENGTH HSPA	C8-C9	C9-C11	C9-C11	C10-C13
CAS NUMBER	64742-48-9 (R)	64742-48-9 (R)	64742-48-9 (R)	64742-48-9 (R)
EC NUMBER	932-020-9	919-857-5	919-857-5	918-481-9

- Paints
- Coatings
- Resins
- Cleaning

Dearomatized aliphatic fluids

TotalEnergies

KETRUL

ALIPHATIC FLUIDS - DEAROMATIZED

Typical values

PROPERTIES	UNITS	METHODS	KETRUL D 70	KETRUL D 80 HN	KETRUL D 85	KETRUL D 100
DENSITY AT 15°C (59°F)	kg/m³	ASTM D4052	817	816	818	815
SAYBOLT COLOR	Unit Saybolt color	ASTM D6045	+30	+30	+30	+30
BOILING RANGE: INITIAL POINT	°C (°F)	ASTM D86	197 (387)	200 (392)	212 (414)	235 (455)
FINAL POINT			238 (460)	237 (459)	242 (468)	262 (504)
FLASH POINT	°C (°F)	ASTM D93	72 (162)	76 (169)	85 (185)	102 (216)
EVAPORATION RATE	Ether=1	DIN 53170	550	800	1000	> 1000
AROMATIC CONTENT	mg/kg (ppm)	UV Internal method	< 30	< 30	< 30	< 30
BENZENE CONTENT	mg/kg (ppm)	ASTM D6229	< 1	< 1	< 1	< 1
SULPHUR CONTENT	mg/kg (ppm)	ASTM D5453	< 1	< 1	< 1	< 1
ANILINE POINT	°C (°F)	ASTM D611	68 (154)	68 (154)	70 (158)	77 (171)
POUR POINT	°C (°F)	ASTM D97	< -50 (-58)	< -50 (-58)	< -50 (-58)	< -45 (-49)
VAPOR PRESSURE AT 20°C (68°F)	kPa	Calculated	0.019	0.018	0.013	0.002
VISCOSITY AT 20°C (68°F)	mm²/s	ASTM D445	2.3	2.4	2.7	3.5
AT 40°C (104°F)			1.6	1.7	1.9	2.3

	KETRUL D 70	KETRUL D 80 HN	KETRUL D 85	KETRUL D 100
CARBON CHAIN LENGTH HSPA	C11-C14	C11-C14	C11-C14	C13-C16
CAS NUMBER	64742-47-8 (R)	64742-47-8 (R)	64742-47-8 (R)	64742-46-7 (R)
EC NUMBER	926-141-6	926-141-6	926-141-6	934-954-2

- Decorative Paints
- Defoamers
- Cleaning

04.

Product overview

Special ranges: Tailored for the needs of the application

HYDROSEAL: Plasticizer for Silicone Sealants

HYDROSEAL

TotalEnergies

- HYDROSEAL acts as a plasticizer in silicone sealants, primarily used in construction, covering both new builds (professional applications) and renovations (mainly DIY).

- It replaces silicone oil to help reduce the costs of the sealant.

- Low aromatic content ensures excellent color stability and minimized odor.

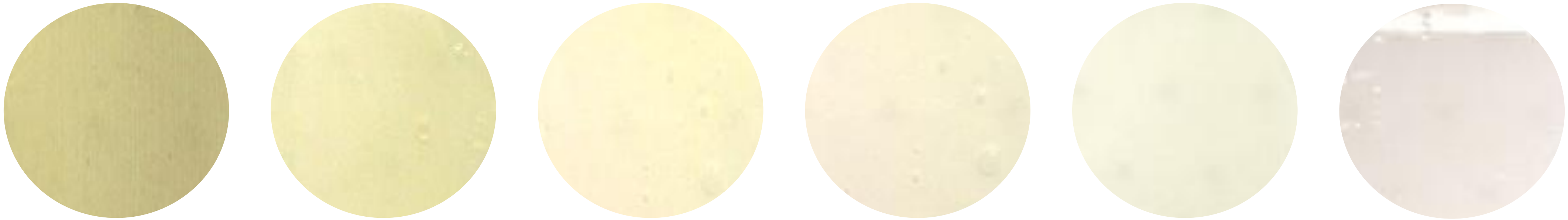
- Depending on the grade, HYDROSEAL can represent between 5% and 50% of the sealant formulation.

Technical support can be provided

- Homologated at all top players in Europe.

- HYDROSEAL brand is the market reference

Influence of aromatics in the plasticizer



1%	0,5%	0,2%	0,1%	0,05%	< 0,03% (< 300 ppm)

Aromatic content in organic plasticizer

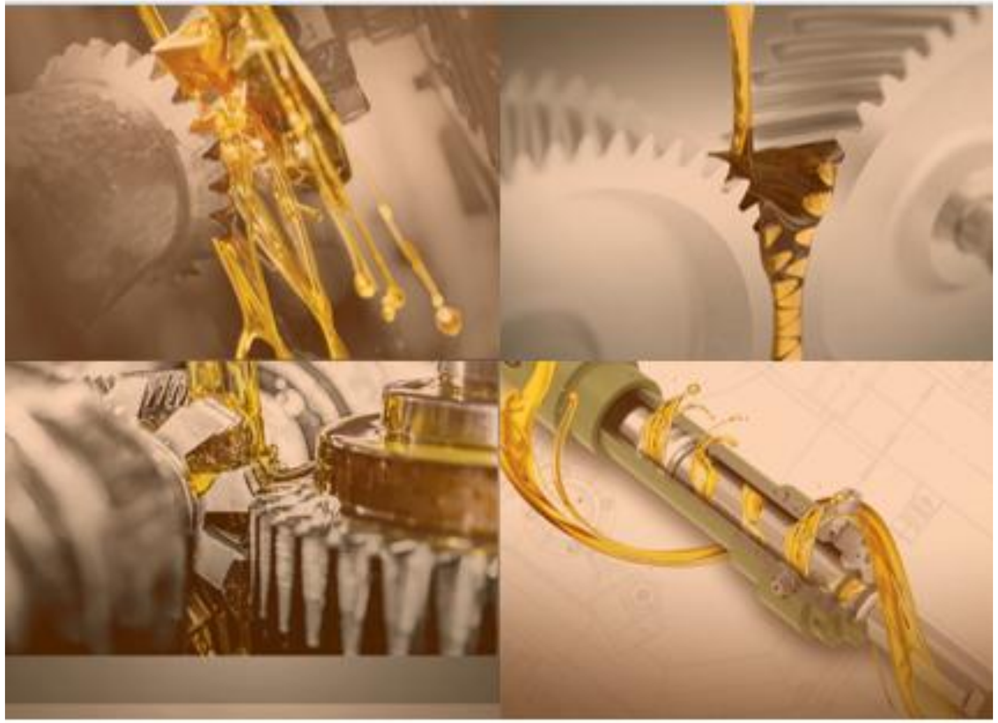
All HYDROSEAL grades contain less than 300 ppm aromatics, No risk of yellow sealants!

HYDROSEAL: Plasticizer for Silicone Sealants

HYDROSEAL acts as a plasticzer in the silicone sealant.

Product	Max. compatibility	Uses
HYDROSEAL G 232 H	> 60%	For DIY (Do It Yourself) sealants. In some countries (e.g. Turkey, Spain) this product is preferred due to the customs tariff number.
HYDROSEAL G 240 H	~ 55%	For DIY (Do It Yourself) sealants.
HYDROSEAL BioLife 260	~ 45%	For DIY (Do It Yourself) sealants and multi-purpose sealing products.
HYDROSEAL G 250 H	~ 40%	For DIY (Do It Yourself) sealants and multi-purpose sealing products.
HYDROSEAL G 3 H	~ 35%	For standard bathroom, glazing and multi-purpose sealing products.
HYDROSEAL BioLife 290	~ 30%	For professional quality sealants.
HYDROSEAL G 400 H	~ 20%	For professional quality sealants.
HYDROSEAL G 340 H	~ 10%	For professional quality sealants.

BERYLANE: Lubricants markets



BERYLANE®

A Full Range of Dearomatized, Carbon-Neutral and Bio-Circular Fluids for Lubricants, Metalworking Fluids & Greases

TotalEnergies

Specialfluids.com

TotalEnergies
a multi-energy company

TotalEnergies is a global integrated energy company that produces and markets energies: oil and biofuels, natural gas, biogas and low-carbon hydrogen, renewables and electricity. We are committed to provide as many people as possible with energy that is more reliable, more affordable and more sustainable. TotalEnergies places sustainability at the heart of its strategy, projects and operations.

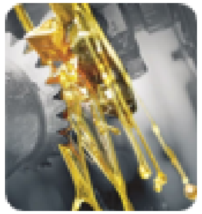
The Special Fluids division of TotalEnergies: **Innovative, Pure and Sustainable Fluids for Today and Tomorrow**

TotalEnergies Fluids is a leader in the design, production and sale of high purity, biodegradable hydrocarbon fluids. Our Hydro De-Aromatization (HDA) technology and narrow distillation cuts allow us to produce high-purity fluids with ultra-low aromatic content and NSF-H1 compliance in our plants in Oudalle, France, and Bayport, USA.

BERYLANE® Hydrocarbon Fluids

At the beginning of every lubricant and fluid formulation stands the crucial choice of the base oils and the additives. To support formulators in creating high-performance products, TotalEnergies Fluids has developed the BERYLANE ranges: BERYLANE, BERYLANE NeutraLife, BERYLANE BioLife.

All ranges are designed to enhance performance, whether used as fluids or as functional additives. They play a central role in ensuring optimal lubrication, thermal stability, and compatibility with other raw materials, while contributing to key properties such as precise viscosity control, excellent low-temperature behavior, and excellent oxidation stability.




BERYLANE®

All BERYLANE fluids are pure, low-aromatic and show excellent ageing stability and very good foaming behavior. Their purity and constant quality make them an excellent choice for your formulation, whether used across a wide range of applications including lubricants, greases, metalworking and rolling oils, corrosion protection, textile processing, or as formulation tools to fine-tune viscosity.

Our BERYLANE® grades guarantee

High Purity Products for a Low HSE Impact

- Compliant with FDA 21 CFR 172.3620(b) and NSF-H1
- Not classified as a substance with acute or chronic toxicity
- Not classified as carcinogenic, mutagenic or reprotoxic (CMR)
- Not classified as flammable according to CLP/ GHS and transportation regulations
- Not classified as a skin irritant or sensitizer

Robust Protection for Machine Tools and Metal Workpieces

- Non-corrosive due to extra low sulfur content
- Does not stain yellow alloys or aluminium
- No degradation such as deposits or unexpected increases in viscosity
- Excellent oxidation and thermal stability

From dearomatized to 100% Bio-based Fluids

Certified Sustainability

Carbon Footprint versus alternative Technologies

In accordance to ISO standards 14040/ 14044 / 14067

BERYLANE fluids are tested and approved by TotalEnergies Lubricants for their products

Key Benefits

BERYLANE®

Proven quality dearomatized hydrocarbon fluids, available in numerous viscosity grades and offering reliable performance across a wide range of applications.

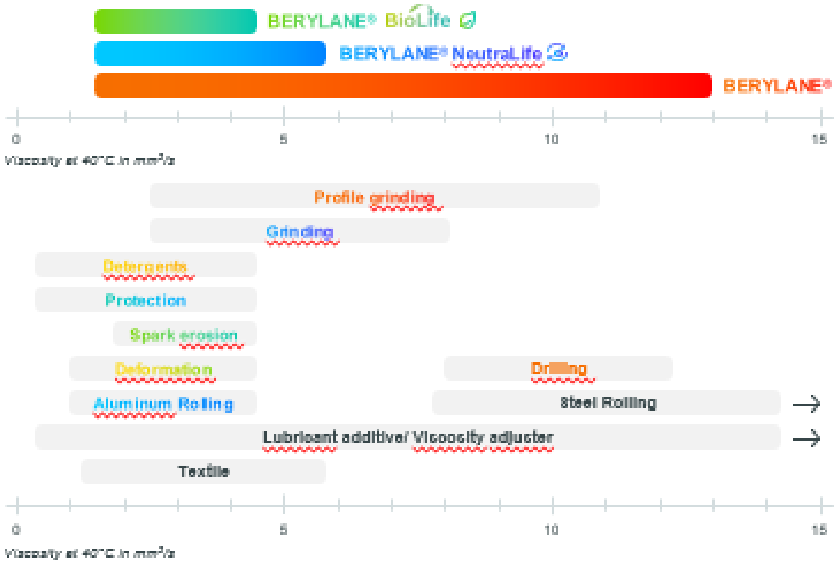
BERYLANE® NeutraLife

Delivers carbon neutrality supported by a partially bio-based composition, without compromising the performance and compatibility of the BERYLANE range.

100% BIO-BASED BERYLANE®

Combines the technical strengths of the BERYLANE range with a 100% bio-based content, offering a negative carbon footprint and an exceptionally low pour point while maintaining performance profile and compatibility. An excellent choice for high-performance, low-temperature applications.

A Broad Range of Viscosity Grades for a Wide Range of Applications



1) ISCCA: The eco-friendly range has also earned ISCC Plus Certification, a well-respected environmental standard for bio-based approaches that verifies compliance with sustainability and traceability criteria throughout the product lifecycle. https://specialfluids.totalenergies.com/en/iscc-plus-certificate

2) TotalEnergies Ecosolutions: BERYLANE BIOLIFE has been awarded the TotalEnergies Ecosolutions label, a TotalEnergies company standard developed in compliance with ISO 14020 and 14021 that recognizes our most efficient solutions in terms of energy efficiency and environmental impact. https://specialfluids.totalenergies.com/en/certifications/our-certifications-and-standards*

**The standards methology and principles have been verified via an audit by an indepdent firm.*

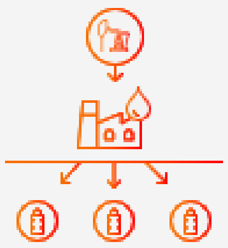
BERYLANE® Ranges: Key properties

Measure	Appearance	Aromatic content	Viscosity at 40°C	Flash Point	Pour Point	Aniline Point
Method	Visual	UV	ASTM D 445	ASTM D 93	ASTM D 97	ASTM D 611
Unit	-	ppm	mm²/s	°C	°C	°C
BERYLANE						
BERYLANE 200	Clear & Bright	<50	1.7	78	-71	70
BERYLANE 230	Clear & Bright	<50	2.4	104	-45	80
BERYLANE 250	Clear & Bright	<50	2.9	118	-36	84
BERYLANE 255	Clear & Bright	<50	3.5	122	-23	89
BERYLANE 275	Clear & Bright	<50	4.1	139	-16	93
BERYLANE 300	Clear & Bright	<150	6.0	160	0	99
BERYLANE 335	Clear & Bright	<150	12.4	184	-24	108
BERYLANE BioLife						
BERYLANE BioLife 25	Clear & Bright	<10	2.6	114	-74	94
BERYLANE BioLife 35	Clear & Bright	<10	3.3	130	-50	99
BERYLANE BioLife 40	Clear & Bright	<10	4.0	148	-39	100
BERYLANE Neutralife						
BERYLANE NeutraLife 3	Clear & Bright	<50	2.6	106	-51	84
BERYLANE NeutraLife 4	Clear & Bright	<50	3.5	122	-27	93
BERYLANE NeutraLife 5	Clear & Bright	<50	5.4	150	-6	100

Choose your **path**

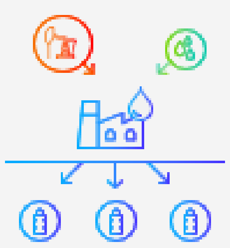
BERYLANE®

100% Fossil
CO_2 positive



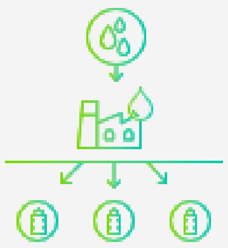
BERYLANE® NeutraLife

Partially Bio-based
CO_2 neutral



BERYLANE® BioLife

Partially Bio-based
CO_2 negative



BERYLANE BioLife : From Bio-circular Materials to Pure Hydrocarbons

Life Cycle Assessment (LCA)

BioLife range has a sustainable biosourced origin with carbon footprint available

Because of the carbon dioxide (CO_2) capture during plant growth, **the GHG emission net impact is even negative**. Biogenic sequestration is superior to emissions during the process of harvesting and transformation. Carbon footprint resulting values depend on feedstock origin starting from **- 2,438** kg CO_2 eq./mt of products.

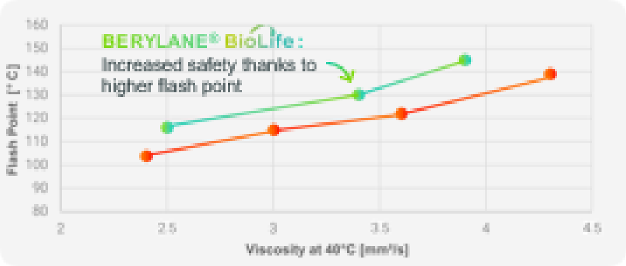
Obtained through ISO 14067, peer reviewed, cradle to gate.

LCA is done in 16 criteras including CO_2 footprint.

Enabling Sustainability Without Compromising On Performance

BERYLANE® BioLife:

High Flash Point
Low Viscosity



BERYLANE® BioLife
BERYLANE®

BERYLANE® BioLife:

Very Low Pourpoint Built for Extreme Conditions

TotalEnergies Fluids

Tour Coupole,
2, place Jean Miller
La Défense 6
92078 - Paris
La Défense Cedex
France

TotalEnergies

Contact us

AQUALANE: Polyacrylamide

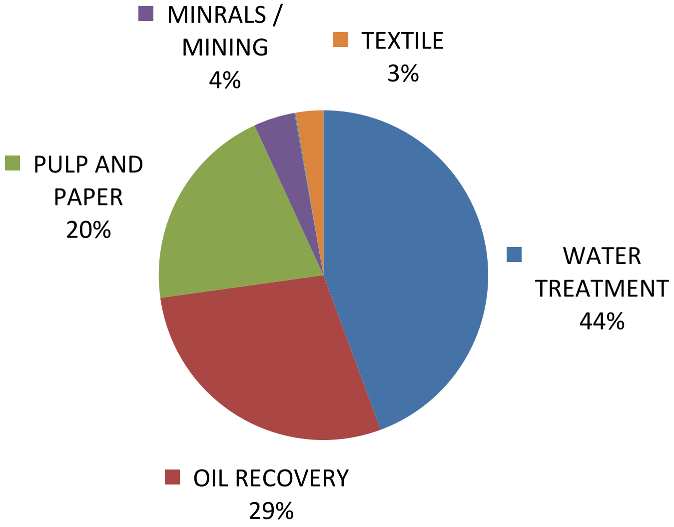
PAM: Main applications

Paper industry
Flocculent

Water treatment
Industrial flocculant for wastewater treatment plants

Oil recovery
Reducing viscosity for fracking operations

Metal separation
Particle separation by size and weight



Textile printing
Increasing the viscosity of inks and water-based systems

AQUALANE: Facts

AQUALANE range fluids are used for production of INVERSE EMULSION POLYACRYLAMIDES (EPAM).

- Main producer (outside China)
  - SNF
  - Ecolab
  - Kemira
  - Solenis
  - Solvay

Nearly all of them are Key accounts, managed from Paris.
(With the exception of SNF, supply is carried out without distributors.)

Main products:

- AQUALANE 75
- AQUALANE 100
- AQUALANE 130

Customers knows exactly which products they need

Cleaning

- Industrial Cleaning
- Semiconductor / Electronics
- Dry Cleaning / Textile Cleaning

Industrial Cleaning

Industrial Cleaning: Main Segments

Metal cleaning

- Aeronautics
- Construction
- Mechanical engineering
- Food industries

Precision cleaning

- Watches
- Printed circuit boards
- Computers,
- Electronics, ...

CLEANING: Our products for Metal and Precision cleaning

Product name	Unit	SOLANE 100-140 HN	SOLANE 100-160	SPIRDANE D 25	SPIRDANE D 40	SPIRDANE D 60	ISANE BIOLIFE 175
Density at 15°C	kg/m³	760	766	769	787	809	757
Saybolt Colour	-	30	30	30	32,12	30	30
Flash Point	°C	11	16	25	43	67	64
Initial Boiling Point	°C	118	124	141	153	188	177
Dry Point	°C	137	156	157	194	212	195
Aromatic content	ppm (m/m)	0	2	1	2	10	0

Success stories: ISANE BioLife 175

DENSO

Hungary

Successful replacement of Exxon ISOPAR L

Industrial parts cleaning in closed cleaning machines

LUBRICANTS. TECHNOLOGY. PEOPLE. FUCHS

Spain

Use as solvent in bio-based industrial cleaner

What are your experiences?

Cleaning: Semiconductor / Electronics

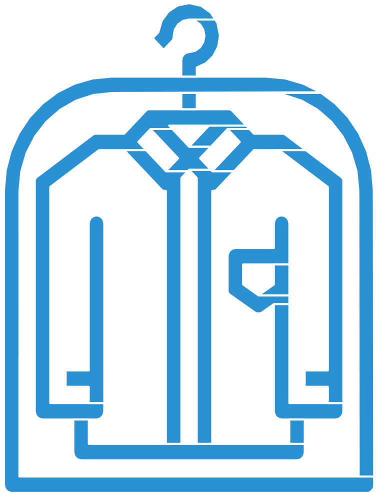
Why choosing our solvents

- Sourced directly from the producer
- No external repacking or blending steps
- Reduced risk of impurities

SOLANE n-HEPTANE HP	ISANE BioLife 120	SOLANE Methylcyclohexane	SPIRDANE D 25
Narrow and precisely controlled boiling range	Isoparaffinic hydrocarbon Nearly Odorless 100% bio-based Highest material compatibility	Higher solvency compared to iso- and n-paraffins Narrow and precisely controlled boiling range	Higher Flashpoint Lower evaporation

Dry Cleaning

DRY CLEANING: DO YOU KNOW THAT?



Dry cleaning

is the cleaning of textiles and clothes, by using a solvent instead of water, while ensuring their preservation.

Multisolvent machines

Multisolvent machines are available.
Certifications specifies which solvent family can be used for each machine.

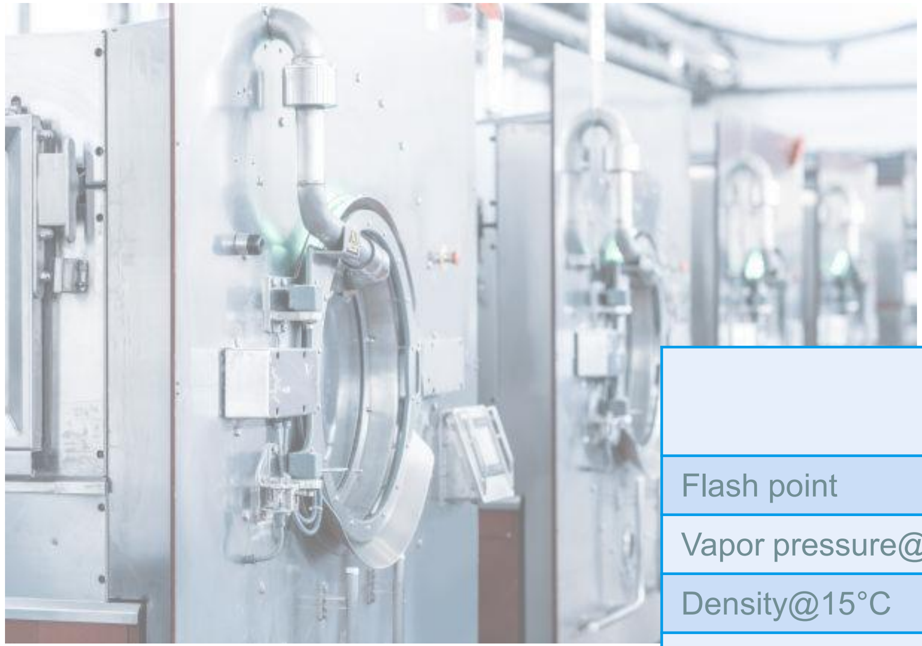
Dry cleaning operations

The main Dry cleaning operations are:

- **Prewashing**
- **Washing**
- Drying
- Finishing (Ironing…)

Washing and drying are carried out in a **closed-circuit** cleaning machine.

Dry cleaning machines



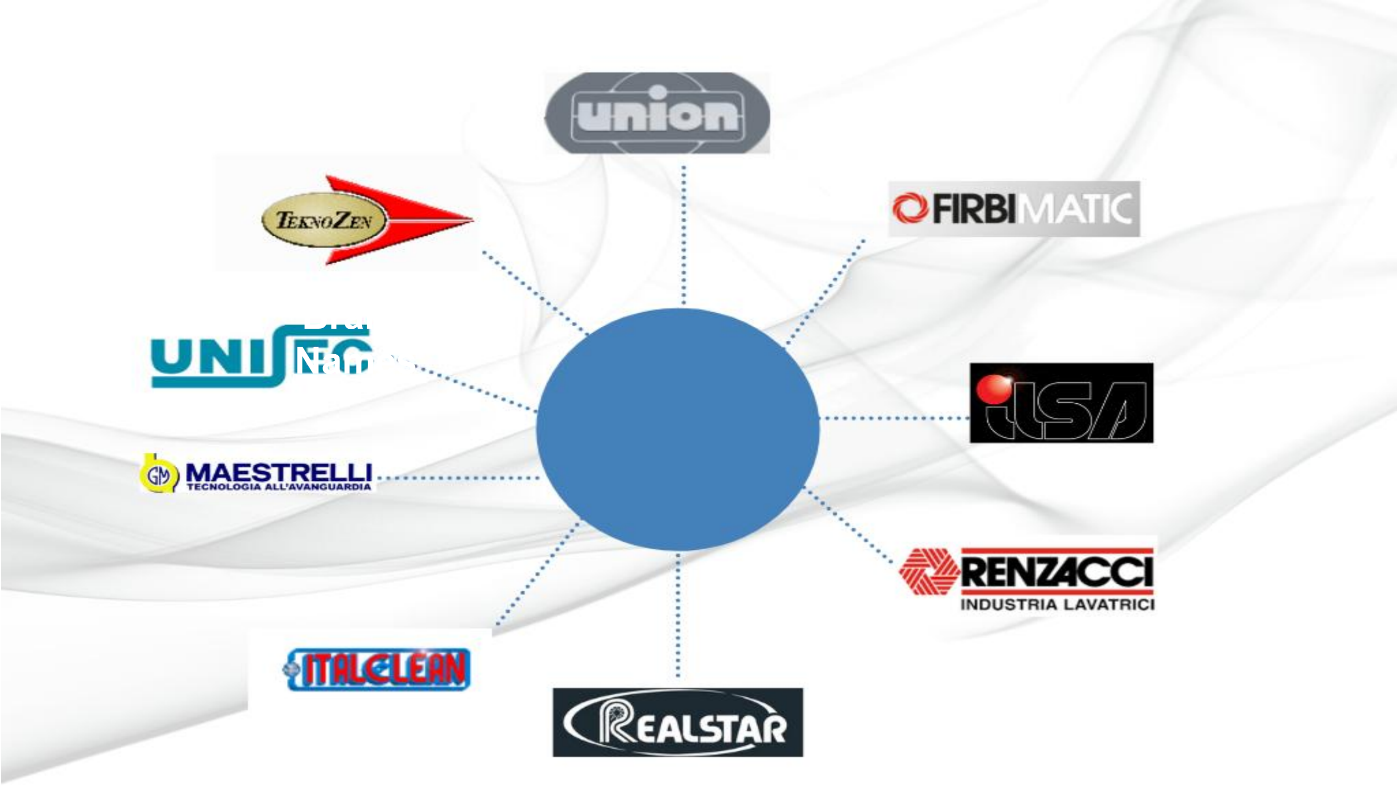
	Unit	ISANE BioLife 175	PERCHLORO-ETHYLENE
Flash point	°C	65	NA
Vapor pressure@20°C	kPa	0,06	1,91
Density@15°C	kg/m3	761	1630
LFL - UFL	%	0.6 - 6.5	NA
VOC	-	100%	100%
Odor	-	No odor	Typically ether
Biodegradable	-	Readily	Not
CMR	-	Not classified	Carc Cat 2
Aquatic chronic toxicity		Not classified	Cat 2

CMR: Carcinogenic, mutagenic, reprotoxic; LFL: Lower Flammability Limit in Air; UFL: Upper Flammability Limit in Air; VOC: Volatile organic compound

DRY CLEANING: Machine producer

TotalEnergies

Specific machine models of these brand names work with pure hydrocarbons solvents



DRY CLEANING: Prewash - For heavy fat and oil stains

	Unit	SOLANE ISOHEXANE
Flash point	°C	-20
Vapor pressure@20°C	kPa	26
Density@15°C	kg/m3	660
LFL - UFL	%	1.2 - 7
VOC	-	100%

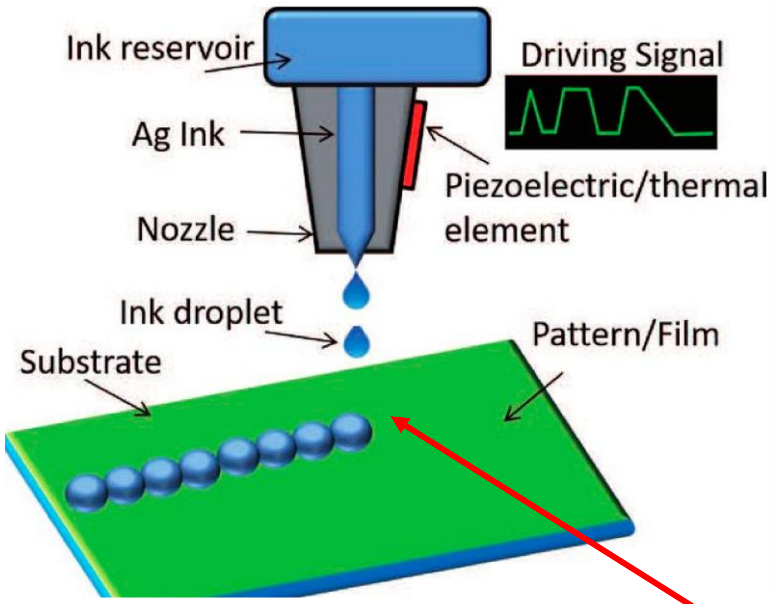
Only for prewash applictions, no use in the machine!

SCRIPTANE CE: Solvents for Ceramic Inks

Printing Inks: Ceramics - Digital printing

TotalEnergies

Process:



Stable viscosity for constant flow properties required

Range nomenclature
Product names describe the viscosity.

SCRIPTANE CE 4= Viscosity at 40°C: 4 mm²/s

Low Aromatic content
Safe use

Low Sulfur content
No staining/corrosion of the printing machine.

Our ranges: **SCRIPTANE CE & SCRIPTANE BioLife**

Thank you